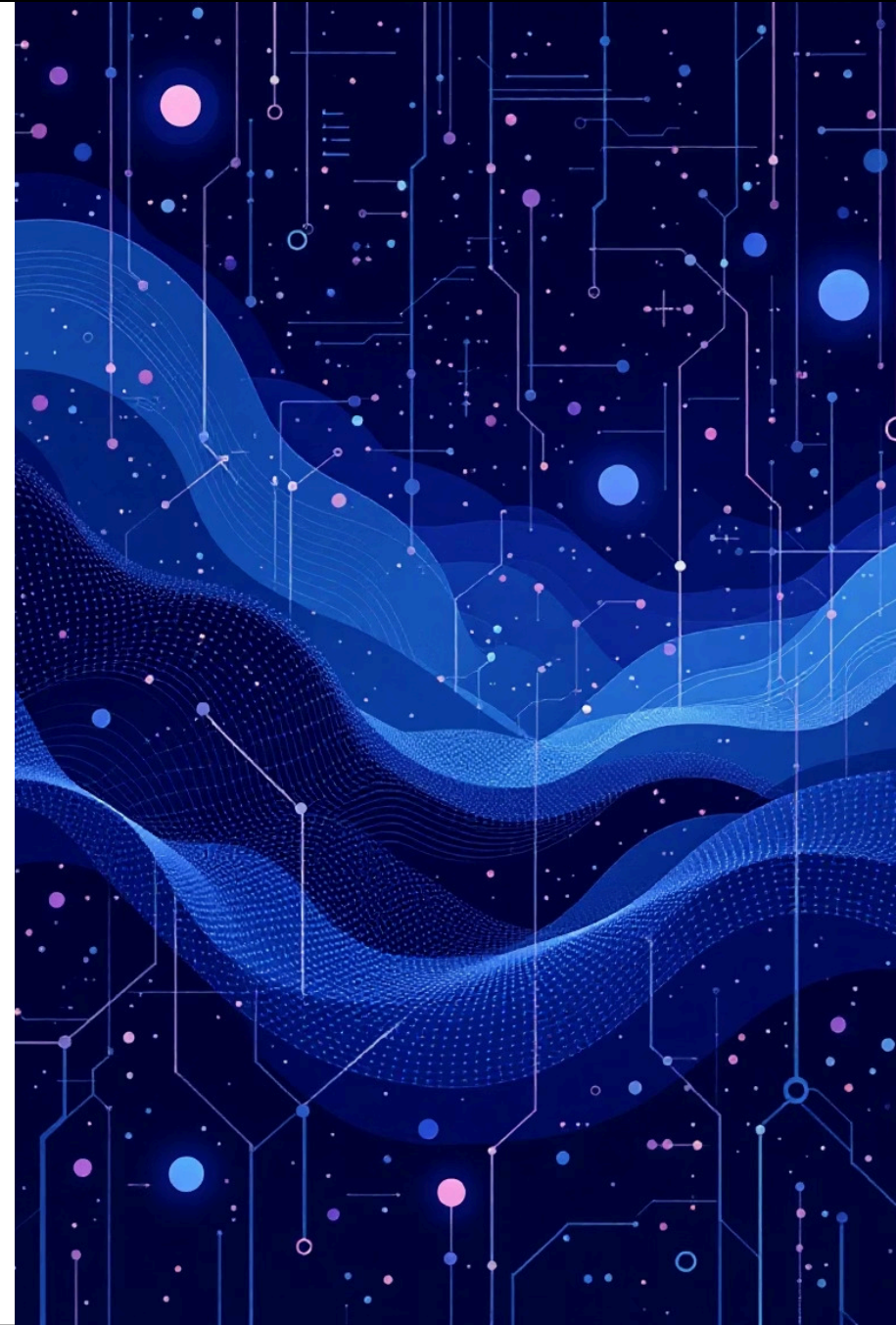


MODULE 01 – Class 1

GenAI, NLP and Prompt Engineering

Generative AI Landscape & Responsible Use



Course Introduction and Today's Focus

Course Subject

GenAI, NLP and Prompt Engineering

Module 01

Generative AI Landscape & Responsible Use

Why This Course Matters Today

Generative AI is transforming how we create content, solve problems, and interact with technology. Understanding its foundations, capabilities, and limitations is essential for professionals across all industries. This course equips you with practical knowledge to leverage AI responsibly and effectively.

What Today's Class Will Cover

We'll explore the fundamentals of Generative AI, distinguish it from traditional AI approaches, understand the evolution of language models, and examine critical concepts like hallucination and responsible use. You'll gain a comprehensive foundation for working with modern AI systems.

What is Generative AI and Why It Is Different

Generative AI represents a fundamental shift in how machines interact with information. Unlike systems that simply analyze or categorize, GenAI creates entirely new content.

Traditional AI

- Focuses on classification and prediction
- Analyzes existing data patterns
- Makes decisions based on training
- Outputs are typically labels or scores

Generative AI

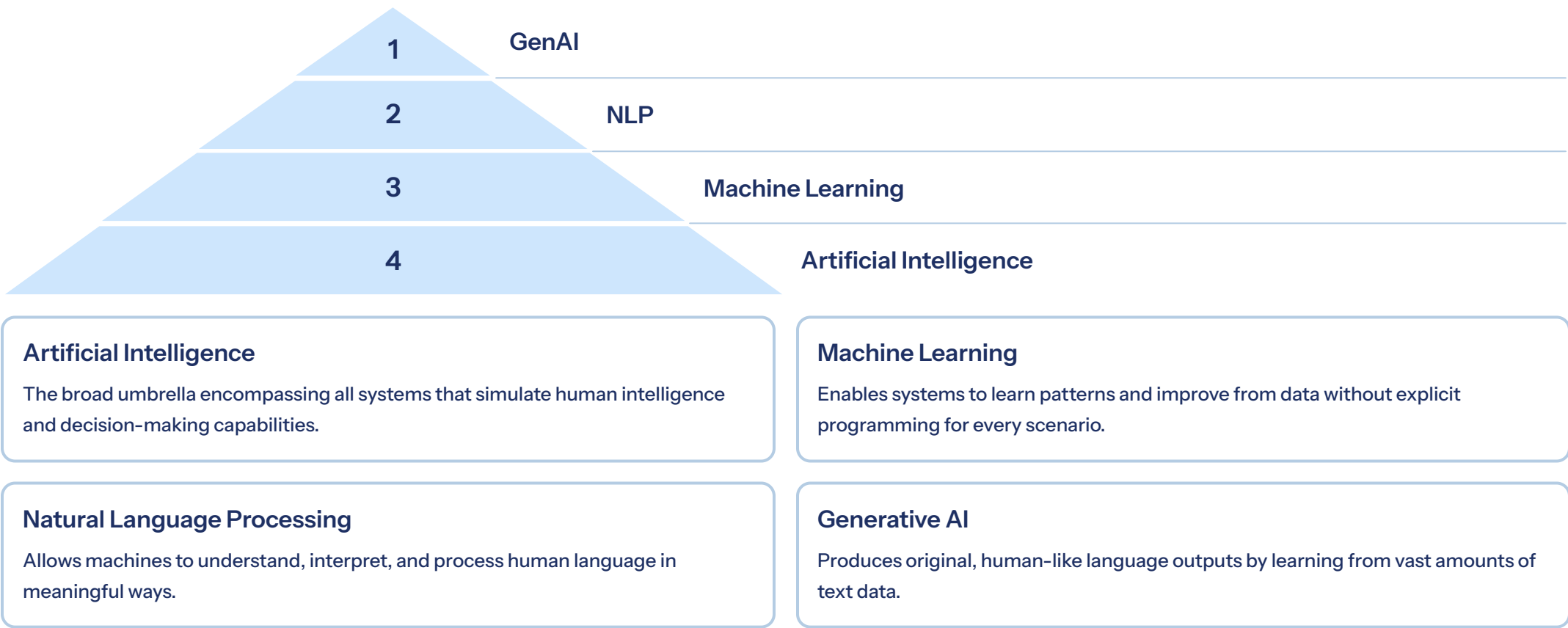
- Creates new, original content
- Generates text, images, code, and summaries
- Produces human-like outputs
- Widely used in education and industry

 **Key Insight:** Responsible use is essential from the beginning. Understanding both capabilities and limitations ensures ethical and effective application of Generative AI.



AI, ML, NLP, and GenAI – Conceptual Relationship

Understanding how these technologies relate to each other provides crucial context for working with modern AI systems. Each layer builds upon the previous one, creating increasingly sophisticated capabilities.



How AI Systems Work

AI systems follow a structured learning process that transforms raw data into intelligent outputs. Understanding this fundamental flow is essential for grasping how both traditional AI and GenAI operate.



Data Collection

Gathering relevant information from diverse sources to build the foundation for learning.



Pattern Learning

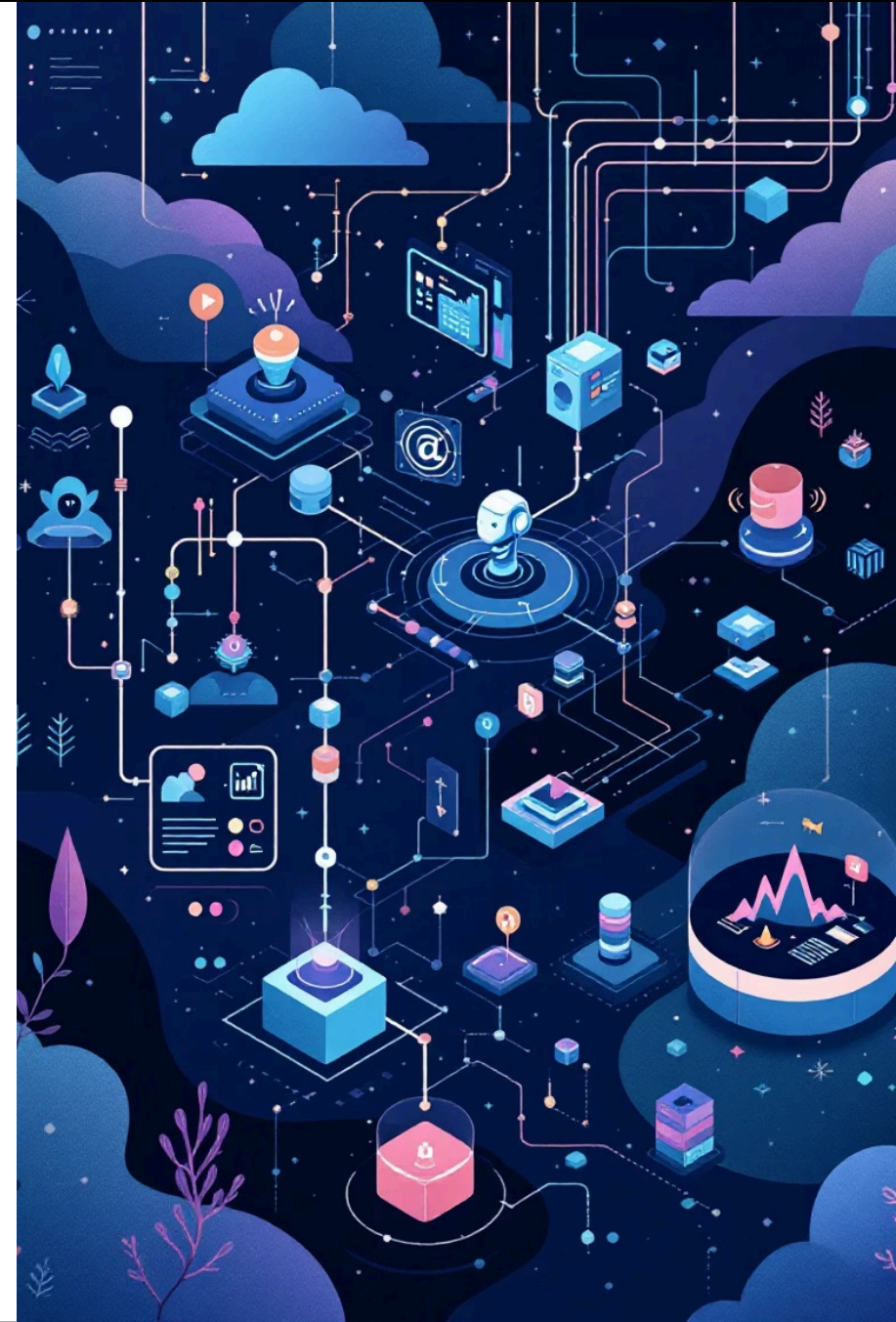
The system analyzes data to identify relationships, trends, and underlying structures.



Task Execution

Applying learned patterns to perform predictions, classifications, or generate new content.

This three-stage operation forms the foundation for all AI and GenAI systems, whether they're recommending products, translating languages, or creating original text.



Evolution of Language Models

Language AI has progressed through distinct technological stages, each building upon the limitations of its predecessors. This evolution has led to the powerful generative systems we use today.

1 Rule-Based Systems

Fixed instructions and predefined patterns. Limited flexibility and required manual updates for new scenarios.

2 Statistical Models

Used probability and frequency analysis. Improved accuracy but still required extensive feature engineering.

3 Neural Models

Learned sequential patterns automatically. Better at understanding context and generating coherent sentences.

4 Transformers

Self-attention mechanisms enabled modern LLMs. Revolutionary improvement in handling long-range context and generating human-like text.



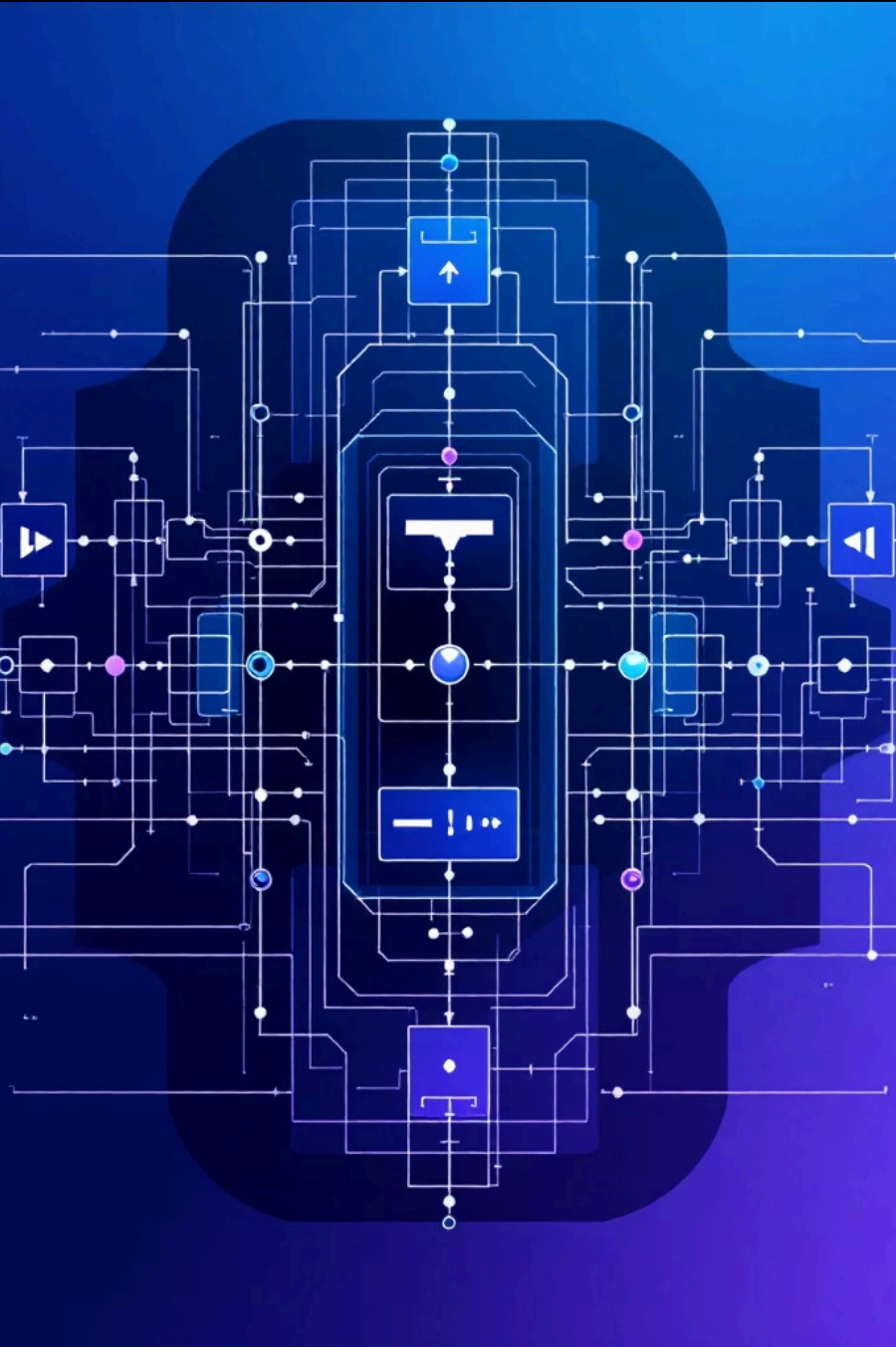
From Rule-Based Systems to Transformers

The journey from rigid rule-based systems to flexible transformers represents one of the most significant technological shifts in AI. Each approach brought unique strengths and limitations.

Rule-Based	Statistical	Neural	Transformers
Fixed instructions	Probability-based	Sequential learning	Self-attention
No real understanding	Feature engineering required	Improved coherence	Long context handling
Manual updates needed	Better classification	Automatic patterns	Human-like generation

The Key Shift

Transformers introduced self-attention mechanisms that allow models to weigh the importance of different words in context, regardless of their position. This breakthrough enabled effective handling of long-range dependencies and nuanced language understanding.



What Are Large Language Models (LLMs)?

Large Language Models represent the cutting edge of Generative AI. These transformer-based systems are trained on massive datasets and generate human-like text by predicting one token at a time.



User Prompt

User provides a question or instruction to the model.



Embedding Conversion

Tokens are converted into numerical vectors that capture semantic meaning.



Probability Scoring

Model calculates probability distributions for potential next tokens.



Response Generation

Process repeats token by token until complete response is generated.



Tokenization

Text is broken into smaller units (tokens) for processing.



Transformer Neural Layers

Deep neural network layers process the embeddings using self-attention mechanisms.



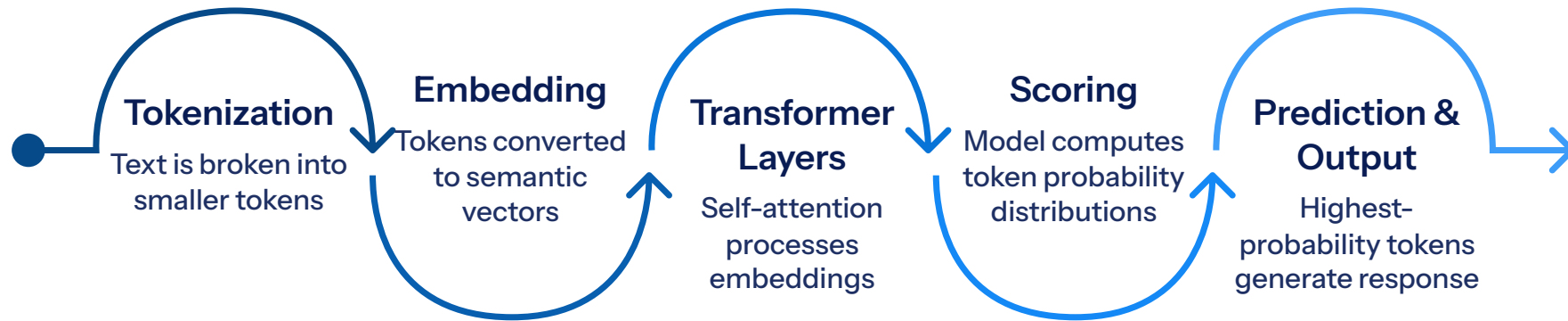
Next Token Prediction

Highest probability token is selected based on the scoring.

 **Important:** LLMs do not store facts like databases. They generate responses based on learned patterns from training data, which means their outputs are predictions, not retrieved information.

Hallucination and Why It Happens

One of the most critical concepts in understanding LLMs is hallucination—when models generate confident but incorrect information. This isn't a bug, it's an inherent characteristic of how these systems work.



Grounding

Connect responses to verified sources and factual databases to reduce false information.



Verification

Always validate critical information from LLM outputs against reliable sources.



Awareness

Understanding that confidence doesn't equal accuracy helps users interpret outputs appropriately.

Risks and Responsible AI Use

While Generative AI offers tremendous potential, it also introduces significant risks that require careful consideration and proactive management. Responsible use begins with understanding these challenges.

Risk Category	Description & Impact
Hallucination	Generation of false or fabricated information presented confidently. Can spread misinformation and erode trust in AI systems.
Bias	Unfair patterns learned from training data that perpetuate stereotypes and discriminatory outcomes across protected categories.
Misuse	Academic dishonesty, impersonation, generating harmful content, and other intentional misapplications of the technology.
Privacy Violations	Inadvertent disclosure of sensitive information, training data leakage, and unauthorized use of personal data.

Mitigation Strategies

- Implement technical controls and safety measures
- Practice ethical prompting and responsible use
- Verify outputs and maintain human oversight
- Follow organizational policies and guidelines

Your Role

Every user plays a part in ensuring AI is used responsibly. By understanding these risks and applying best practices, you contribute to safer, more beneficial AI deployment across education and industry.



Debrief, Reflection, and Homework

Debrief & Class Sharing

- What did you learn about how Generative AI works?
- One key difference between traditional AI and GenAI
- One risk you found most important (hallucination, bias, or misuse)
- Share one example of a safe and unsafe AI prompt
- Why responsible AI use is important in academics

Homework

- Write 5 safe prompts for a campus chatbot (public information only)
- Identify 3 unsafe prompts and rewrite them ethically
- Read short note on AI → ML → NLP → GenAI progression
- Be prepared to discuss prompt structure in next class

Thank You

Great work completing Module 01! You now understand the foundations of Generative AI, how language models work, and the importance of responsible AI use.

Coming Up Next

Module 02: Tokens, Embeddings & Vector Semantics

Understanding how machines transform language into mathematical representations they can process and compare

